# Reading Note: Score Centering in Off-Policy RL

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**Derivation map.**

$$\text{policy gradient} \longrightarrow \text{off-policy bias} \overset{\text{controlled experiment}}{\longrightarrow} \text{trainer drift},$$
$$\underbrace{\text{exact IS}}_{\text{unbiased, high variance}} \longrightarrow \underbrace{\text{TIS/MIS}}_{\text{lower variance, biased}} \longrightarrow \text{score centering}.$$

## 0 Notation and Policy Gradient

We follow the notation of Marek and Ryabinin [1]. Let $p_\theta$ be the trainer policy and $q_\theta$ the sampler policy. For a rollout $y = (y_1, \dots, y_T)$,

$$J(\theta) \coloneqq \mathbb{E}_{y\sim p_\theta}[R(y)], \qquad S_\theta(y) \coloneqq \nabla_\theta \log p_\theta(y) = \sum_{t=1}^{T} \nabla_\theta \log p_\theta(y_t \mid y_{<t}). \tag{1}$$

**Note: Score function**

$$S_\theta(y) = \nabla_\theta \log p_\theta(y).$$

$S_\theta(y)$ is the gradient of a single sample w.r.t. the parameters, i.e., it points toward the parameter change that most rapidly increases the likelihood of $y$—hence the name *score function*.

**Policy gradient theorem.** Assume that $R$ has no direct dependence on $\theta$. Define

$$s_{y_t} \coloneqq \nabla_\theta \log p_\theta(y_t \mid y_{<t}), \qquad Q_t^{p_\theta}(y_{\le t}) \coloneqq \mathbb{E}_{y_{>t}\sim p_\theta(\cdot|y_{\le t})}[R(y)]. \tag{2}$$

Then

$$\boxed{\begin{aligned}\nabla_\theta J(\theta) &= \mathbb{E}_{y\sim p_\theta}[R(y) S_\theta(y)] \\ &= \mathbb{E}_{y\sim p_\theta}\left[\sum_{t=1}^{T} Q_t^{p_\theta}(y_{\le t}) s_{y_t}\right].\end{aligned}} \tag{3}$$

*Proof sketch.*

$$\begin{aligned}
\nabla_\theta J(\theta) &= \sum_y R(y) \nabla_\theta p_\theta(y) \\
&= \mathbb{E}_{p_\theta}[R(y) \nabla_\theta \log p_\theta(y)] \\
&= \sum_{t=1}^{T} \mathbb{E}_{p_\theta}[R(y) s_{y_t}] \\
&= \sum_{t=1}^{T} \mathbb{E}_{p_\theta}[\mathbb{E}_{p_\theta}[R(y) \mid y_{\le t}] s_{y_t}] \\
&= \mathbb{E}_{p_\theta}\left[\sum_{t=1}^{T} Q_t^{p_\theta}(y_{\le t}) s_{y_t}\right].
\end{aligned}$$

**Note: Policy gradient theorem**

The theorem writes the gradient of expected return as an expectation of return-weighted policy scores. Its derivation uses the standard *log-derivative trick*:

$$\nabla_\theta \log p_\theta(y) = \frac{\nabla_\theta p_\theta(y)}{p_\theta(y)} \quad \Longleftrightarrow \quad \nabla_\theta p_\theta(y) = p_\theta(y) \nabla_\theta \log p_\theta(y).$$

Fix one prefix $y_{<t}$. For $v \in \mathcal{V}$, define

$$p_v \coloneqq p_\theta(v \mid y_{<t}), \qquad q_v \coloneqq q_\theta(v \mid y_{<t}), \qquad s_v \coloneqq \nabla_\theta \log p_v. \tag{4}$$

The score identity is

$$\mathbb{E}_p[s_{y_t}] = \sum_{v \in \mathcal{V}} p_v s_v = \sum_{v \in \mathcal{V}} p_v \nabla_\theta \log p_v = \nabla_\theta \sum_{v \in \mathcal{V}} p_v = 0. \tag{5}$$

All expectations below are conditional on the fixed prefix; the conditioning is suppressed to keep the equations compact.

# 1 Systematic Drift in Off-Policy Updates

Off-policy sampling generally biases the policy-gradient estimator:

$$\mathcal{B}(q, p) \coloneqq \mathbb{E}_q[R s_{y_t}] - \mathbb{E}_p[R s_{y_t}] \neq 0. \tag{6}$$

The key question is

*Which component enters the online feedback loop and causes instability?*

## 1.1 Conditioned Consistency of SFT and Offline RL

For binary rewards $R \in \{0, 1\}$, define the rejection-sampled distribution

$$q_0^{\mathrm{rej}}(x, y) \coloneqq q_0(x, y \mid R = 1). \tag{7}$$

Then

$$\begin{aligned} g_{\mathrm{RL}}^{\mathrm{rej}} &\coloneqq \mathbb{E}_{(x,y)\sim q_0^{\mathrm{rej}}}\left[R(x,y)\nabla_\theta \log p_\theta(y \mid x)\right] \\ &= \mathbb{E}_{(x,y)\sim q_0^{\mathrm{rej}}}\left[\nabla_\theta \log p_\theta(y \mid x)\right], \\ g_{\mathrm{SFT}} &\coloneqq \mathbb{E}_{(x,y)\sim q_0^{\mathrm{rej}}}\left[\nabla_\theta \log p_\theta(y \mid x)\right]. \end{aligned} \tag{8}$$

Therefore,

$$g_{\mathrm{RL}}^{\mathrm{rej}} = g_{\mathrm{SFT}}. \tag{9}$$

**Insight: Stable Offline but Unstable Online.**

SFT and rejection-sampled offline RL have identical gradients, SFT remains stable under a large sampler–trainer gap while online RL can be unstable under a much smaller gap, *which leaves online updating as the prime suspect.*

### 1.2 The Curse of Trainer Drift

Motivated by the contrast behavior of SFT and offline RL, Marek and Ryabinin [1] perturb the trainer's weights to create sampler and compare offline and online RL training with binary rewards, where only $R = 1$ rollouts contribute to the gradient.

$$p_{\theta_t}\ (\text{trainer}) \xrightarrow{\text{sync+perturb}} q_t\ (\text{sampler}) \xrightarrow[R=1\ \text{only}]{\text{rollouts}} g_t \xrightarrow{\text{update}} p_{\theta_{t+1}}.$$

And they find the following *abnormal pattern*,

| | offline | online under TIM |
|---|---|---|
| +1/0 | stable | least stable |
| group-centered | unstable | most stable |

The central empirical insight:

- The same positive-only update is stable when the teacher is fixed but unstable when the teacher is refreshed from the student;
- Reducing the mean reward substantially mitigates this instability.

The comparison isolates two factors:

$$\underbrace{p_{\theta_t} \longrightarrow q_t \longrightarrow p_{\theta_{t+1}}}_{\text{moving teacher}} \qquad \text{and} \qquad \underbrace{\mu_q \coloneqq \mathbb{E}_q[R]}_{\text{mean training weight}} .$$

The two factors play different roles. The moving teacher is an amplifier: it feeds any systematic per-step error into the next trainer. The unresolved question is whether a nonzero $\mu_q$ can itself produce such an error under mismatch. To test this, retain the mean weight while removing all task signal: assign the same reward $c$ to every rollout,

$$R(y) \equiv c, \qquad \mu_q = \mathbb{E}_q[R] = c. \tag{10}$$

At a fixed prefix, define the mean score under the sampler $q$:

$$\bar{s} \coloneqq \mathbb{E}_q[s_{y_t}] = \sum_v q_v \nabla_\theta \log p_v. \tag{11}$$

Because every rollout receives the same reward, the true gradient is zero. On policy,

$$J(\theta) = c, \qquad \nabla_\theta J(\theta) = 0, \qquad \mathbb{E}_p[c\, s_{y_t}] = c\nabla_\theta \sum_v p_v = 0. \tag{12}$$

Under mismatch, the token is sampled from $q$ while the score is computed through $p$:

$$\mathbb{E}_q[c\, s_{y_t}] = c\bar{s}, \qquad \bar{s} = \sum_v q_v \nabla_\theta \log p_v \neq \sum_v p_v \nabla_\theta \log p_v = 0. \tag{13}$$

$$\underbrace{\mathbb{E}_q[s_{y_t}]}_{\text{off-policy}} = \bar{s} \neq 0 \qquad \text{whereas} \qquad \underbrace{\mathbb{E}_p[s_{y_t}]}_{\text{on-policy}} = 0$$

**This nonzero off-policy score expectation is the source of drift.** Multiplying it by the mean training weight gives the systematic bias $c\bar{s} \neq 0$, which can accumulate under a moving teacher.

### 1.3 Pinpointing the Source of Instability

Let $\mu_q := \mathbb{E}_q[R]$ and decompose

$$R = \mu_q + (R - \mu_q). \tag{14}$$

The off-policy update is

$$\begin{aligned} \underbrace{\mathbb{E}_q[R s_{y_t}]}_{\text{off-policy update}} &= \mathbb{E}_q\big[\big(\mu_q + (R-\mu_q)\big) s_{y_t}\big] \\ &= \mu_q \bar{s} + \mathbb{E}_q[(R-\mu_q) s_{y_t}] \\ &= \mu_q \bar{s} + \mathbb{E}_q[(R-\mu_q)(s_{y_t} - \bar{s})] + \underbrace{\mathbb{E}_q[R-\mu_q]}_{0} \bar{s} \\ &= \underbrace{\mu_q \bar{s}}_{\text{drift}} + \underbrace{\mathrm{Cov}_q(R, s_{y_t})}_{\text{signal}}. \end{aligned} \tag{15}$$

On policy, $\mathbb{E}_p[s_{y_t}] = 0$, so

$$\mathbb{E}_p[R s_{y_t}] = \mathrm{Cov}_p(R, s_{y_t}). \tag{16}$$

Therefore, the full off-policy bias in equation (6) is

$$\mathcal{B}(q,p) = \underbrace{\mu_q \bar{s}}_{\text{trainer drift}} + \underbrace{\mathrm{Cov}_q(R, s_{y_t}) - \mathrm{Cov}_p(R, s_{y_t})}_{\text{remaining distributional bias}}. \tag{17}$$

The drift term is then singled out because it survives without task signal and has a systematic direction. Treating $q$ as stop-gradient,

$$\bar{s} = \sum_v q_v \nabla_\theta \log p_v = -\nabla_\theta \underbrace{\left( -\sum_v q_v \log p_v \right)}_{\mathcal{L}_{\mathrm{CE}}(q,p)}. \tag{18}$$

For $\mu_q > 0$, the drift minimizes $\mathcal{L}_{\mathrm{CE}}(q,p)$, distilling the sampler $q$ (teacher) into the trainer $p$ (student).

1. **Feedback amplification.**

$$\underbrace{p_{\theta_t}}_{\text{trainer}} \xrightarrow{\text{refresh}} \underbrace{q_t}_{\text{sampler}} \xrightarrow{\mu_{q_t}\bar{s}_t} \underbrace{p_{\theta_{t+1}}}_{\text{trainer}}.$$

The drift distills the trainer toward the sampler; refreshing the sampler closes the loop and accumulates this bias.

2. **Mean-reward control.**

$$\mu_q \to 0 \quad \Longrightarrow \quad \|\mu_q \bar{s}\|_2 \to 0.$$

This explains the greater online stability of group-relative rewards and joint positive–negative training over positive-only training.

## 2 Exact Importance Sampling

Assume $p_\theta(y) > 0 \Rightarrow q_\theta(y) > 0$ and define the sequence ratio

$$r(y) := \frac{p_\theta(y)}{q_\theta(y)} = \prod_{t=1}^{T} \frac{p_\theta(y_t \mid y_{<t})}{q_\theta(y_t \mid y_{<t})}. \tag{19}$$

Then

$$\mathbb{E}_{q_\theta}[r(y)R(y)S_\theta(y)] = \sum_y q_\theta(y)\frac{p_\theta(y)}{q_\theta(y)}R(y)S_\theta(y) = \mathbb{E}_{p_\theta}[R(y)S_\theta(y)] = \nabla_\theta J(\theta). \tag{20}$$

Hence exact sequence-level IS is unbiased.

At the fixed prefix, define the token ratio

$$r_v := \frac{p_v}{q_v}. \tag{21}$$

Its weighted score has zero mean:

$$\mathbb{E}_q[r_{y_t} s_{y_t}] = \sum_v q_v r_v s_v = \sum_v p_v s_v = 0. \tag{22}$$

### 2.1 Why exact IS has high variance

The total coordinate-wise variance is

$$\begin{aligned} \operatorname{tr}\operatorname{Cov}_{q_\theta}\big(r(y)R(y)S_\theta(y)\big) &= \mathbb{E}_{q_\theta}[\|r(y)R(y)S_\theta(y)\|^2] - \|\mathbb{E}_{q_\theta}[r(y)R(y)S_\theta(y)]\|^2, \\ \mathbb{E}_{q_\theta}[\|r(y)R(y)S_\theta(y)\|^2] &= \sum_y \frac{p_\theta(y)^2}{q_\theta(y)} R(y)^2 \|S_\theta(y)\|^2. \end{aligned} \tag{23}$$

If $q_\theta(y) \approx 0$ while $p_\theta(y) > 0$, the second moment is large. Consequently, updates on tokens with low sampler probability are particularly variance-prone and unstable, as also observed empirically in DPPO [7]. Moreover,

$$\log r(y) = \sum_{t=1}^{T} \log \frac{p_\theta(y_t \mid y_{<t})}{q_\theta(y_t \mid y_{<t})}, \tag{24}$$

so sequence length compounds token-level mismatch [3, 4].

Exact IS: unbiased + heavy-tailed variance

**Note: Token-level ratios are not exact sequence IS.** Let

$$r_t := \frac{p_\theta(y_t \mid y_{<t})}{q_\theta(y_t \mid y_{<t})}, \qquad r(y) = \prod_{t=1}^{T} r_t.$$

Then

$$\mathbb{E}_q\left[R(y)\sum_t r_t s_{y_t}\right] \neq \mathbb{E}_q\left[r(y)R(y)\sum_t s_{y_t}\right] = \mathbb{E}_p\left[R(y)\sum_t s_{y_t}\right]$$

in general. The token-level estimator corrects only the current-token distribution, not the prefix and continuation distributions; it therefore does not inherit the unbiasedness of exact sequence IS.

## 3 TIS and MIS: Variance–Bias Trade-off

Exact IS can assign an unbounded weight to a token unlikely under the sampler:

$$\|Rr_v s_v\|_2 = |R|\frac{p_v}{q_v}\|s_v\|_2 \longrightarrow \infty \qquad (q_v \to 0,\ p_v > 0).$$

Such rare updates can dominate the estimator variance.

- **Truncated Importance Sampling (TIS):** $f_{\text{TIS}}(r_v) = \min(r_v, c)$.
- **Masked Importance Sampling (MIS):** $f_{\text{MIS}}(r_v) = r_v \mathbf{1}\{\ell \le r_v \le u\}$.

Here $c > 0$ and $0 < \ell < u$ are thresholds. Both trade unbiasedness for lower tail variance and more stable updates.

TIS and MIS: lower variance + biased

## 4 From Residual Drift to Score Centering

This section we answer:

*Why focus on score?*

At a fixed prefix, the residual drift can be removed by centering either factor of the original update:

$$\underbrace{\mathbb{E}_q[(R - \mathbb{E}_q[R])s_{y_t}]}_{\text{reward centering}} = \text{Cov}_q(R, s_{y_t}) = \underbrace{\mathbb{E}_q[R\,(s_{y_t} - \mathbb{E}_q[s_{y_t}])]}_{\text{score centering}}. \tag{25}$$

**Why center scores?** [1]

- **No critic:** token-level value estimation is hard.
- **Deterministic additive correction:** given the prefix, $\bar{s}$ does not depend on the sampled token and can be added directly to the policy loss.
- **Top-$k$ implementation:** only sampler top-$k$ logprobs are needed with the paper's tail approximation; runs with $k = 128$ finish within 1% of the baseline wall-clock time

in the reported experiments.

### 4.1 Vanilla score centering

The paper therefore defines

$$\boxed{\widetilde{s}_{y_t} = s_{y_t} - \bar{s} = s_{y_t} - \sum_v q_v s_v}. \tag{26}$$

Therefore

$$\mathbb{E}_q[R\widetilde{s}_{y_t}] = \mathrm{Cov}_q(R, s_{y_t}). \tag{27}$$

### 4.2 Score centering after TIS or MIS

For $w_v \coloneqq f(r_v)$, center the reweighted score in the same way:

$$\boxed{R\left(w_{y_t} s_{y_t} - \mathbb{E}_q[w_{y_t} s_{y_t}]\right) = R\left(w_{y_t} s_{y_t} - \sum_v q_v w_v s_v\right)}. \tag{28}$$

## 5 Practice of Score Centering

The full-vocabulary score-centering loss is

$$L_{\mathrm{SC}} \coloneqq -R\left(\log p_{y_t} - \sum_{v\in\mathcal{V}} \mathrm{sg}[q_v] \log p_v\right). \tag{29}$$

Its ascent direction is

$$-\nabla_\theta L_{\mathrm{SC}} = R\left(\nabla_\theta \log p_{y_t} - \sum_v q_v \nabla_\theta \log p_v\right) = R(s_{y_t} - \bar{s}). \tag{30}$$

Storing the sampler's full next-token distribution at every position is prohibitively expensive. Let $H$ contain its top-$k$ tokens and let $\rho$ be the ratio of sampler to trainer probability mass outside $H$:

$$\rho \coloneqq \frac{1 - \sum_{v\in H} q_v}{1 - \sum_{v\in H} p_v}.$$

The paper uses the top-$k$ loss

$$\widehat{L}_{\mathrm{SC}} = -R\left(\log p_{y_t} - \sum_{v\in H} \mathrm{sg}[q_v - \rho p_v] \log p_v\right). \tag{31}$$

Here sg denotes stop-gradient [1].

## 6 Summary

$$
\begin{aligned}
\text{on-policy:}\quad & \mathbb{E}_p[s_{y_t}] = 0, \qquad \mathbb{E}_p[R s_{y_t}] = \mathrm{Cov}_p(R, s_{y_t}),\\
\text{off-policy:}\quad & \mathbb{E}_q[s_{y_t}] = \bar{s} \neq 0,\\
& \mathbb{E}_q[R s_{y_t}] = \mathbb{E}_q[R]\bar{s} + \mathrm{Cov}_q(R, s_{y_t}),\\
\text{exact IS:}\quad & \mathbb{E}_{q_\theta}[r(y)R(y)S_\theta(y)] = \mathbb{E}_{p_\theta}[R(y)S_\theta(y)],\\
& \mathbb{E}_{q_\theta}[\|r(y)R(y)S_\theta(y)\|_2^2] = \sum_y \underbrace{\frac{p_\theta(y)^2}{q_\theta(y)}}_{\to\infty \text{ as } q_\theta(y)\to 0} R(y)^2 \|S_\theta(y)\|_2^2,\\
\text{TIS/MIS:}\quad & f(r) \neq r \implies \mathrm{Bias}_f = \mathbb{E}_q[R(f(r_{y_t}) - r_{y_t}) s_{y_t}] \neq 0 \quad \text{in general,}\\
\text{top-}k\text{ score centering:}\quad & \widetilde{s}^{(k)}_{y_t} = s_{y_t} - \sum_{v \in H} (q_v - \rho p_v) s_v.
\end{aligned}
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